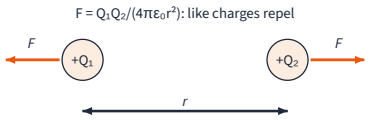
CHAPTER 13 ELECTRO-STATICS

Key idea: charges exert forces through electric fields; potential is the work done per unit charge.

1 COULOMB'S LAW



$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$$

- $\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9$ N m² C⁻²
- Like charges repel, unlike attract
- In a medium: divide by ε_r
- Several charges: add forces as **vectors**

2 ELECTRIC FIELD

$$E = \frac{F}{q} \qquad E = \frac{Q}{4\pi\varepsilon_0 r^2} \qquad E = \frac{V}{d}$$

- Force per unit **positive** charge; N C⁻¹ = V m⁻¹
- Lines start on +, end on −; never cross

3 GAUSS'S LAW

$$\Phi = \oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

Charge distribution	E
Sphere, $r \geq R$	$\frac{Q}{4\pi\varepsilon_0 r^2}$
Infinite line	$\frac{\lambda}{2\pi\varepsilon_0 r}$
Infinite sheet	$\frac{\sigma}{2\varepsilon_0}$
Conductor surface	$\frac{\sigma}{\varepsilon_0}$

Inside a conductor $E = 0$; charge sits on the surface.

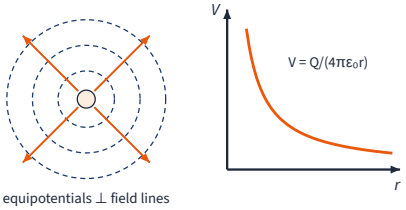
4 ELECTRIC POTENTIAL

$$V = \frac{Q}{4\pi\varepsilon_0 r} \qquad U = qV$$

$$U = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r}$$

- Work done per unit positive charge from **infinity**
- Scalar: add potentials algebraically
- Work by external force: $W = q\Delta V$

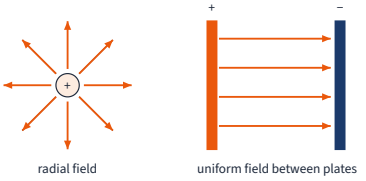
5 FIELD AND POTENTIAL



$$E = -\frac{dV}{dr}$$

- E points towards **lower** potential
- Equipotentials ⊥ field lines; no work along them
- Close equipotentials = strong field

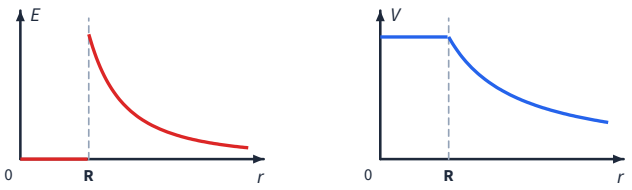
6 CHARGES IN A UNIFORM FIELD



$$a = \frac{qE}{m} \qquad \tfrac{1}{2}mv^2 = qV$$

- Enters ⊥ to the field: **parabolic** path
- Oil drop at rest: $\frac{qV}{d} = mg$

7 CHARGED CONDUCTING SPHERE



Inside: $E = 0$, V constant. Outside: like a point charge at the centre.

8 COMMON SECTION A TRAPS

- Dropping the sign of a negative charge in V (keep it; V is a scalar).
- Adding field magnitudes without directions.
- Using the diameter for r, or µC without ×10⁻⁶.
- $E = 0$ does **not** mean $V = 0$.
- Halving r: F ×4, V ×2.

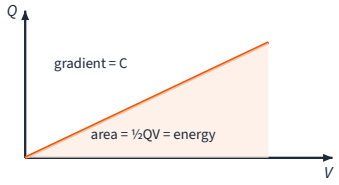
EXAM TIPS

- ✓Draw each field vector before adding.
- ✓Use energy, $\frac{1}{2}mv^2 = qV$, for speeds.
- ✓State the Gaussian surface you choose.
- ✓Give direction of E as well as magnitude.

CHAPTER 14 CAPACITORS

Key idea: a capacitor stores charge and energy in the electric field between its plates.

1 CAPACITANCE



$$C = \frac{Q}{V}$$

- Unit farad: 1 F = 1 C V^{-1} (μF, nF, pF)
- Gradient of Q–V graph $= C$
- Isolated sphere: $C = 4\pi\varepsilon_0 R$

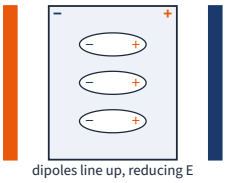
2 PARALLEL PLATES

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d}$$

$$E = \frac{V}{d} = \frac{\sigma}{\varepsilon_0}$$

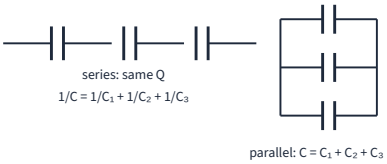
- **Battery connected:** V constant
- **Isolated:** Q constant

3 DIELECTRICS



Insert dielectric	Battery on	Isolated
C	$\times \varepsilon_r$	$\times \varepsilon_r$
Q	$\times \varepsilon_r$	same
V, E	same	$\div \varepsilon_r$
U	$\times \varepsilon_r$	$\div \varepsilon_r$

4 SERIES & PARALLEL



$$C_{\parallel} = C_1 + C_2$$

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$$

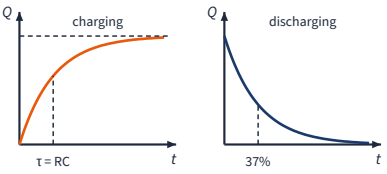
- Series: **same** Q; smaller C takes larger V
- Parallel: **same** V; charges add

5 ENERGY STORED

$$U = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = \frac{Q^2}{2C}$$

- U = area under the V–Q graph
- Energy density $u = \tfrac{1}{2}\varepsilon_0 E^2$
- Battery gives QV: half stored, half lost as heat
- Sharing charge: Q conserved, energy lost

6 CHARGING & DISCHARGING



$$Q = Q_0 e^{-t/RC}$$

$$Q = Q_0\left(1 - e^{-t/RC}\right)$$

- $\tau = RC$: 37% left (discharge), 63% reached (charge)
- $t_{1/2} = RC \ln 2$
- Current decays as $I_0 e^{-t/RC}$ in **both** cases

7 UNIT CHECK

Quantity	Formula	Unit
Capacitance	Q/V	F
Permittivity	ε_0	F m^{-1}
Time constant	RC	s
Energy density	$\frac{1}{2}\varepsilon_0 E^2$	J m^{-3}

8 COMMON SECTION A TRAPS

- Using resistor rules for capacitors (they are reversed).
- Forgetting whether Q or V stays constant.
- μF → ×10^{-6} F; cm^2 → ×10^{-4} m^2.
- 63% is for charging, 37% for discharging.
- Ln graph of discharge: gradient $= -1/RC$.

EXAM TIPS

- ✓Decide first: battery connected or isolated?
- ✓For series, find the common Q first.
- ✓Use $\ln$ to find t from an exponential.
- ✓Quote τ before sketching charge–time graphs.

CHAPTER 15 ELECTRIC CURRENT

Key idea: current is the drift of charge carriers; resistance depends on the material and its shape.

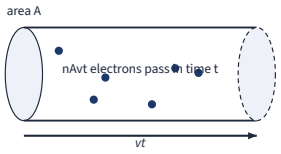
1 ELECTRIC CURRENT



$$I = \frac{dQ}{dt} \qquad Q = Ne$$

- Conventional current: direction of + charge flow
- Electrons drift the opposite way
- 1 A = 1 C s^{-1}

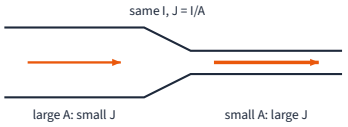
2 DRIFT VELOCITY



$$I = nAev_d$$

- $v_d \sim 10^{-4}$ m s^{-1}; random speed $\sim 10^5$–10^6 m s^{-1}
- Lamp lights at once: the **field** travels at nearly c
- Narrower wire: larger v_d

3 CURRENT DENSITY



$$J = \frac{I}{A} = nev_d \qquad J = \sigma E$$

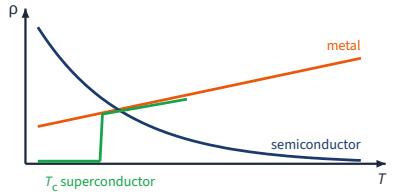
Vector, A m^{-2}; $J = \sigma E$ is the microscopic Ohm's law.

4 RESISTANCE & RESISTIVITY

$$R = \frac{V}{I} \qquad R = \frac{\rho L}{A} \qquad \sigma = \frac{1}{\rho}$$

- ρ (Ω m) and σ are properties of the **material**
- Stretched to k × length (same volume): $R \times k^2$

5 EFFECT OF TEMPERATURE



- **Metal:** ions vibrate more, more collisions, so ρ rises
- **Semiconductor:** more carriers released, so ρ falls
- **Superconductor:** $\rho = 0$ below the critical temperature

6 MICROSCOPIC VIEW

$$\sigma = \frac{ne^2\tau}{m} \qquad v_d = \frac{eE\tau}{m}$$

τ: mean time between collisions (relaxation time).

7 ELECTRICAL POWER

$$P = VI \qquad P = I^2R \qquad P = \frac{V^2}{R}$$

Energy $W = VIt$; 1 kW h = 3.6 × 10^6 J. Heater coil: combine $P = V^2/R$ with $R = \rho L/A$.

8 COMMON SECTION A TRAPS

- Using the diameter as radius in $A = \pi r^2$.
- Thinking drift speed is the speed of the signal.
- Same current, smaller area: J and v_d larger.
- Resistivity does not change when the wire is cut.

EXAM TIPS

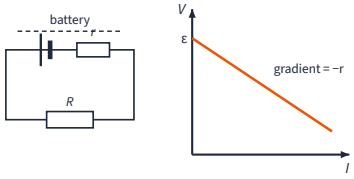
- ✓Write $I = nAev_d$ and rearrange before substituting.
- ✓Use ratios for wires of the same material.
- ✓Explain temperature effects with collisions and carriers.
- ✓Units: n in m^{-3}, A in m^2.

CHAPTER 16

DIRECT CURRENT CIRCUITS

Key idea: charge is conserved at junctions and energy around loops.

1 E.M.F. & INTERNAL RESISTANCE

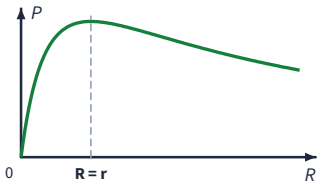


$$E = V + Ir$$

$$E = I(R + r)$$

E.m.f.: energy converted to electrical energy per unit charge. V–I graph: intercept E, gradient $-r$.

2 POWER TRANSFER

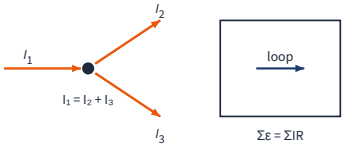


$$P = \frac{E^2 R}{(R+r)^2}$$

$$P_{\max} = \frac{E^2}{4r}$$

Maximum power in R when $R = r$ (efficiency 50%).

3 KIRCHHOFF'S LAWS



$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

$$\sum E = \sum IR$$

- 1st law: conservation of **charge**
- 2nd law: conservation of **energy**
- Negative answer: current flows the other way

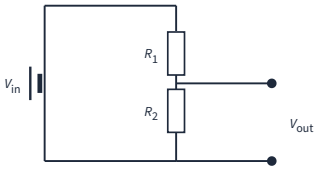
4 RESISTOR NETWORKS

$$R_s = R_1 + R_2$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

- Series: same I; parallel: same V
- Voltmeter across R: $\dfrac{RR_V}{R + R_V}$ (reads low)
- Cells in series: add E and r

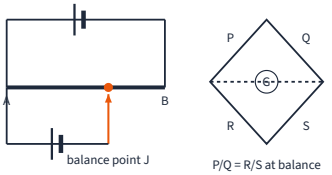
5 POTENTIAL DIVIDER



$$V_{\text{out}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}$$

- Load across R_2: output falls
- Thermistor (NTC): $R \downarrow$ as $T \uparrow$; LDR: $R \downarrow$ as light $\uparrow$

6 POTENTIOMETER & BRIDGE



$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$\frac{P}{Q} = \frac{R}{S}$$

At balance no current in the galvanometer, so the potentiometer reads the true e.m.f.

7 METHOD FOR NETWORKS

1. Label every current with a direction.
2. Junction rule at each node.
3. Loop rule round enough loops, with signs.
4. Solve the simultaneous equations; check one loop.

8 COMMON SECTION A TRAPS

- Taking the terminal p.d. as the e.m.f. when current flows.
- Forgetting r in $I = E/(R + r)$.
- Sign of an e.m.f. met from + to − in a loop.
- Balance length is measured from the **same** end as the driver's positive.

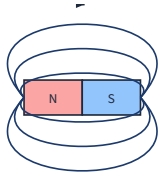
EXAM TIPS

- ✓Redraw the circuit simply before calculating.
- ✓State "at balance no current flows through G".
- ✓Use ratios of lengths for potentiometers.
- ✓Check power: $EI = I^2R + I^2r$.

CHAPTER 17 MAGNETIC FIELDS

Key idea: moving charges and currents create magnetic fields and feel forces in them.

1 MAGNETIC FIELD

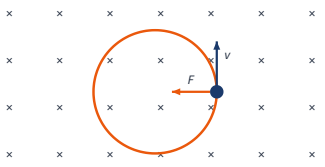


field lines from N to S outside the magnet

$$B = \frac{F}{IL\sin\theta}$$

- 1 T = 1 N A^{-1} m^{-1}
- Lines: closed loops, N → S outside a magnet
- Right-hand grip rule for wires and coils

2 FORCE ON A MOVING CHARGE



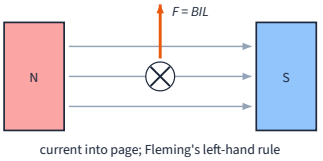
B into page: F = qvB towards centre, r = mv/qB

$$F = qvB\sin\theta$$

$$r = \frac{mv}{qB} \qquad T = \frac{2\pi m}{qB}$$

Force $\perp v$: does no work, speed constant. T independent of v.

3 FORCE ON A CONDUCTOR



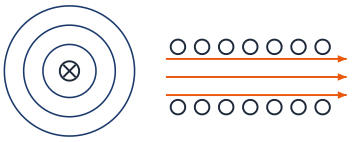
current into page; Fleming's left-hand rule

$$F = BIL\sin\theta$$

$$\tau = NBIA\sin\phi$$

Fleming's left-hand rule: first finger = field, second finger = current, thumb = force.

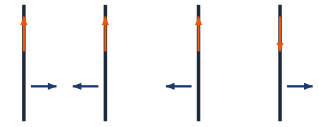
4 FIELDS DUE TO CURRENTS


wire: B = μ₀I/2πr solenoid: B = μ₀nI (uniform inside)

Source	B
Long straight wire	$\frac{\mu_0 I}{2\pi r}$
Centre of coil	$\frac{\mu_0 NI}{2r}$
Long solenoid	$\mu_0 nI$

$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}; n = turns per metre.

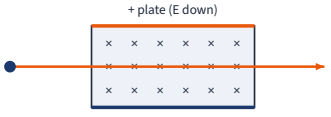
5 PARALLEL WIRES


same direction: attract opposite: repel

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

- Same direction: **attract**; opposite: repel
- Equal and opposite forces (3rd law)
- Defines the ampere: 2×10^{-7} N m^{-1} at 1 m

6 E/M & VELOCITY SELECTOR



B into page
undeflected when eE = evB → v = E/B

$$v = \frac{E}{B} \qquad \frac{e}{m} = \frac{2V}{B^2 r^2}$$

Crossed fields: undeflected when $qE = qvB$. $e/m = 1.76 \times 10^{11}$ C kg^{-1}.

7 HALL EFFECT

$$V_H = \frac{BI}{nte}$$

- Carriers pushed to one side until $eE_H = evB$
- Sign of V_H shows the sign of the carriers
- Semiconductor: small n, large V_H (Hall probe)

8 COMMON SECTION A TRAPS

- For electrons, reverse the left-hand rule.
- Using n as total turns in $B = \mu_0 nI$.
- Saying the magnetic force changes the speed.
- t in V_H is the thickness **along B**.
- Angle θ is between the wire (or v) and B.

EXAM TIPS

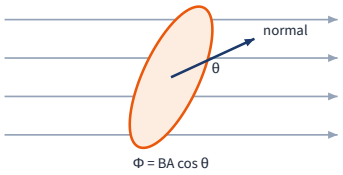
- ✓Draw B, I (or v) and F as a 3-D sketch.
- ✓Write "magnetic force provides centripetal force".
- ✓Add fields from two wires as vectors.
- ✓Check units: T, A, m.

CHAPTER 18

ELECTRO-MAGNETIC INDUCTION

Key idea: a changing flux linkage induces an e.m.f. that opposes the change.

1 MAGNETIC FLUX

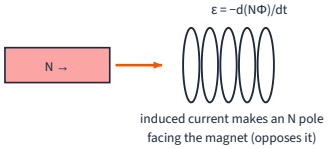


$$\Phi = BA\cos\theta$$

$$\text{linkage} = N\Phi$$

θ between B and the **normal**. 1 Wb = 1 T m².

2 FARADAY'S & LENZ'S LAWS



$$\varepsilon = -N\frac{d\Phi}{dt}$$

- Faraday: $\varepsilon \propto$ rate of change of flux linkage
- Lenz: induced current **opposes** the change (energy conservation)

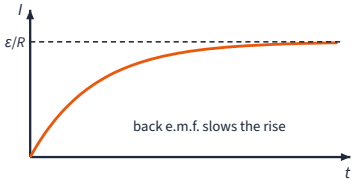
3 MOTIONAL E.M.F.

$$\varepsilon = BLv \qquad F = \frac{B^2L^2v}{R}$$

$$\varepsilon = NBA\omega\sin\omega t$$

Rotating coil (a.c. generator): peak $\varepsilon_0 = NBA\omega$ when the plane is **parallel** to B.

4 SELF-INDUCTION

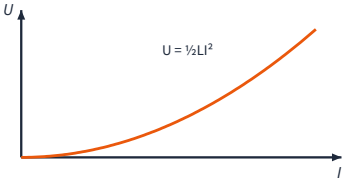


$$\varepsilon = -L\frac{dI}{dt}$$

$$L = \frac{N\Phi}{I} = \frac{\mu_0 N^2 A}{l}$$

Back e.m.f. slows the growth of current; spark when switched off.

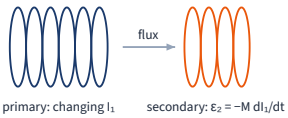
5 ENERGY IN AN INDUCTOR



$$U = \tfrac{1}{2}LI^2 \qquad u = \frac{B^2}{2\mu_0}$$

Stored in the **magnetic field**; returned when the current falls.

6 MUTUAL INDUCTION



$$\varepsilon_2 = -M\frac{dI_1}{dt}$$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

Transformer works on a.c. only; ideal: $V_pI_p = V_sI_s$.

7 UNIT CHECK

Quantity	Unit	In base form
Flux Φ	Wb	T m² = V s
Flux density B	T	Wb m^{-2}
Inductance L, M	H	Wb A^{-1} = V s A^{-1}
E.m.f.	V	Wb s^{-1}

8 COMMON SECTION A TRAPS

- Taking θ from the plane instead of the normal.
- Forgetting N in flux linkage.
- Peak e.m.f. occurs when the flux is **zero**.
- Saying an e.m.f. needs a closed circuit (only a current does).
- Transformer on d.c.: no output.

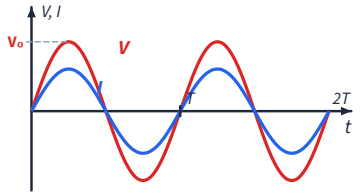
EXAM TIPS

- ✓Find $\Delta(N\Phi)$ first, then divide by Δt.
- ✓Explain direction with Lenz's law in words.
- ✓Use $\varepsilon = BLv$ only when B, L, v are perpendicular.
- ✓Energy: work against the opposing force = electrical energy.

CHAPTER 19 ALTERNATING CURRENT

Key idea: the current and voltage vary sinusoidally and reverse direction every half cycle.

1 BASIC EQUATIONS



Pure resistor: V and I in phase

$$I = I_0 \sin \omega t \qquad V = V_0 \sin \omega t$$

I_0, V_0 peak values

$\omega = 2\pi f$ (rad s^{-1})

$T = \frac{1}{f}$ (s)

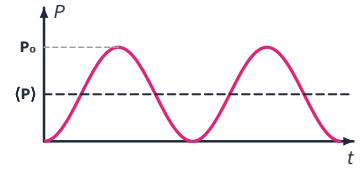
2 R.M.S. (EFFECTIVE) VALUE

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \qquad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

- The steady d.c. that dissipates heat in a resistor **at the same rate** as the a.c.
- Mean value over one cycle $= 0$
- Mains 240 V is r.m.s.: $V_0 = 339$ V

3 POWER IN A.C.

$$\langle P \rangle = I_{\text{rms}} V_{\text{rms}} = I_{\text{rms}}^2 R = \tfrac{1}{2} I_0 V_0$$



- Peak power $P_0 = I_0^2 R = 2\langle P \rangle$
- Instantaneous P varies at **twice** the supply frequency, never negative

Pure L and pure C: mean power $= 0$. Energy is stored, then returned to the source.

4 COMPONENTS

RESISTOR R

- V and I **in phase**, $\phi = 0$
- R independent of f
- Dissipates energy as heat

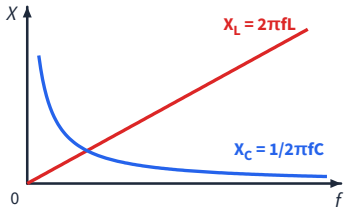
INDUCTOR L

- $X_L = \omega L = 2\pi f L$
- V **leads** I by 90°
- d.c.: $X_L = 0$

CAPACITOR C

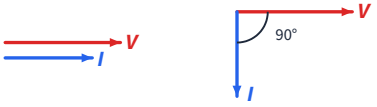
- $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
- I **leads** V by 90°
- Blocks d.c.: $X_C \to \infty$

5 REACTANCE AND FREQUENCY

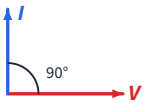


- $X_L \propto f$: straight line through origin, gradient $2\pi L$
- $X_C \propto \frac{1}{f}$: falls towards zero, never touches the axis
- f doubled: X_L doubles, X_C halves

6 PHASE (PHASORS)



R: in phase L: I lags V

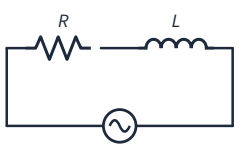


C: I leads V

Series R–L or R–C: phase angle $0 < \phi < 90^\circ$

ELI the ICE man: in L, e.m.f. leads I; in C, I leads e.m.f.

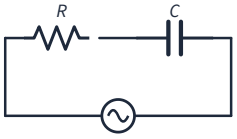
7 R-L AND R-C SERIES CIRCUITS



$$Z = \sqrt{R^2 + X_L^2}$$

$$\tan \phi = \frac{X_L}{R}$$

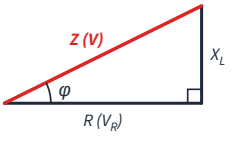
R–L: current **lags** voltage by ϕ



$$Z = \sqrt{R^2 + X_C^2}$$

$$\tan \phi = \frac{X_C}{R}$$

R–C: current **leads** voltage by ϕ



$$V^2 = V_R^2 + V_L^2$$

$$\cos \phi = \frac{R}{Z}$$

$V_{\text{rms}} = I_{\text{rms}} Z$ and mean power $\langle P \rangle = I_{\text{rms}} V_{\text{rms}} \cos \phi = I_{\text{rms}}^2 R$ (only R dissipates power).

8 UNIT CHECK

Quantity	Formula	Unit
Period	$T = 1/f$	s
Angular frequency	$\omega = 2\pi f$	rad s^{-1}
R.m.s. current	$I_{\text{rms}} = I_0/\sqrt{2}$	A
Inductive reactance	$X_L = \omega L$	Ω
Capacitive reactance	$X_C = 1/\omega C$	Ω
Impedance	$Z = \sqrt{R^2 + X^2}$	Ω
Power factor	$\cos \phi = R/Z$	no unit
Mean power	$I_{\text{rms}}^2 R$	W

9 KEY POINTS TO MEMORISE

1. R.m.s. value = peak value $\div \sqrt{2}$ (sine wave only).
2. Only the resistor dissipates mean power.
3. Inductor stores energy in its **magnetic** field; capacitor in its **electric** field.
4. Voltages across R and X are 90° apart: add as **phasors**, not arithmetically.

10 COMMON SECTION A TRAPS

- Reading ω (314 rad s^{-1}) as the frequency: $f = \omega/2\pi = 50$ Hz.
- Using V_0 in $P = V^2/R$ gives the **peak** power, twice the mean.
- Forgetting μF → ×10^{-6} F or mH → ×10^{-3} H.
- Stating that current leads voltage in an inductor (that is the capacitor).

EXAM TIPS

- ✓ Convert peak values to r.m.s. before finding power.
- ✓ Find X_L or X_C first, then Z, then I.
- ✓ Draw the phasor (impedance) triangle: R across, X up, Z hypotenuse.
- ✓ State the phase relation in words: "current leads/lags voltage by …".